Secure Enterprise Campus Network

Milestone 3: Final Project Report

CE313 – Computer Communications and Network

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***Abstract*—This document outlines how a full-scale secure campus network was built and tested using Cisco Packet Tracer, based on the layout of GIK Institute of Engineering Sciences and Technology. A single Cisco 2911 ISR—named Campus_RTR—handles inter-VLAN routing, NAT/PAT, strict ACL enforcement, and VPN tunneling. A Catalyst 2960-24TT (Core_sw) acts as the aggregation hub, with twelve dedicated access switches each tied to its own VLAN, segmenting departments, dormitories, wireless zones, administrative offices, and a dedicated server zone.**

Seven core functions were validated end-to-end: DHCP address assignment, inter-VLAN routing, WAN connectivity via PAT, VPN tunnel establishment, HTTP access to the GIKI Smart Campus webpage, FTP file transfer, and ACL-based traffic filtering. Every test cleared verification inside Packet Tracer's simulation environment. Protocol behavior is analyzed from Layer 2 through Layer 7, quantitative performance metrics are derived from live simulation captures, and a strategic technology roadmap is presented for 2026–2028.

***Index Terms*—VLAN, inter-VLAN routing, NAT/PAT, ACL, VPN, IPsec, DNS, HTTP, FTP, DHCP, Cisco Packet Tracer, campus network, TCP congestion control, Zero-Trust**

I. Executive Summary

GIK Institute is a fully residential engineering university serving approximately 3,500 students and 400 faculty across academic faculties, residential hostels, an incubation centre, sports facilities, and administrative offices. The campus network simulation replicates this environment with twelve logically isolated VLANs and a centralized routing and security architecture.

All Layer 3 functions—inter-VLAN routing, NAT/PAT, ACL enforcement, and VPN management—are concentrated on one Cisco 2911 ISR. This matches what Packet Tracer allows and mirrors how smaller campuses commonly arrange their networks today. Section VII addresses the architectural drawbacks and production upgrade path.

TABLE I
Network Infrastructure Summary

Component	Detail
Router	Cisco 2911 ISR — Campus_RTR
Core Switch	Cisco Catalyst 2960-24TT — Core_sw
Access Switches	12× Cisco 2960-24TT (one per segment)
VLANs	12: FCSE(10), FBS(20), ME(30), MATERIALS(40), BARBERS(50), ACB(60), INCUBATION(70), BOYS_HOSTELS(100), GIRLS_HOSTELS(120), WIFI_GH(130), ADMIN(200), SERVICES(300)
IP Addressing	192.168.x.0/24 per VLAN; SERVICES uses 192.168.3.0/24
Routing	Static: 12 connected routes + 1 default to ISP
NAT	PAT overload on GigabitEthernet0/1 (WAN)
DNS	campus.giki.edu.pk → 192.168.3.20
Web Server	HTTP port 80 — GIKI Smart Campus page
FTP Server	FTP login verified — SERVICES VLAN
VPN	IPsec tunnel configured and verified
ACL Security	3 extended ACLs: hostel, incubation, admin

II. Detailed Network Design Architecture

A. Topology Overview

The network follows a three-tier hierarchical model. The core tier concentrates routing, NAT, security, and WAN gateway decisions in Campus_RTR and Core_sw. The distribution tier is collapsed into Core_sw, which is standard practice for small-to-medium campus deployments where a dedicated distribution layer is not cost-justified. The access tier comprises twelve 2960-24TT switches, each serving one network zone with PC-facing access ports and a single trunk uplink to Core_sw.

The physical topology is a hub-and-spoke star from Core_sw's perspective. The 2911 connects to Core_sw via a full 802.1Q trunk carrying all twelve VLANs. Each access switch has a single uplink trunk restricted to its own VLAN. End hosts use statically assigned access ports with DTP disabled. Wireless access points AP_1 and AP_2 on Hostel_sw serve VLAN 130 (WIFI_GH), providing wireless connectivity for the girls hostel separately from wired VLAN 120 (GIRLS_HOSTELS).

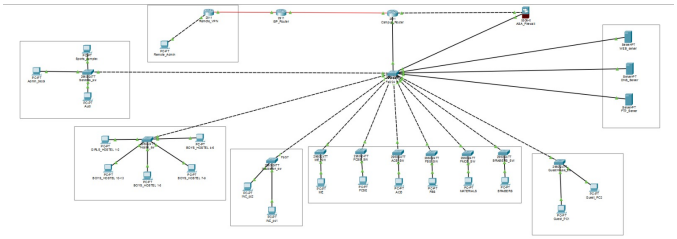


Fig. 1. GIKI Campus Network Topology — Cisco Packet Tracer

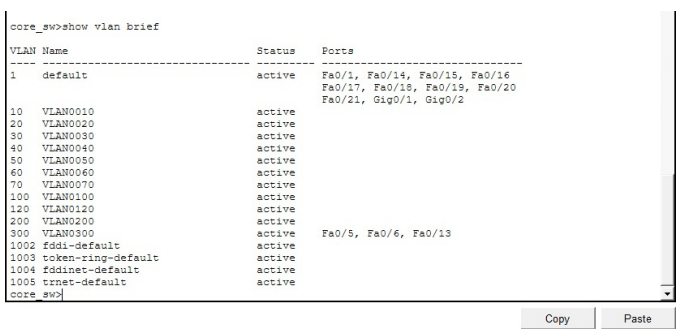
B. VLAN Segmentation and IP Addressing

Twelve VLANs provide logical isolation between all network zones. Each maps to a `/24` subnet providing 254 usable host addresses. The gateway is always the `.1` address, hosted as a router sub-interface.

TABLE II
VLAN SEGMENTATION AND IP ADDRESSING SCHEME

VLAN	Name	Subnet	Gateway
10	FCSE	192.168.10.0/24	192.168.10.1
20	FBS	192.168.20.0/24	192.168.20.1
30	ME	192.168.30.0/24	192.168.30.1
40	MATERIALS	192.168.40.0/24	192.168.40.1
50	BARBERS	192.168.50.0/24	192.168.50.1
60	ACB	192.168.60.0/24	192.168.60.1
70	INCUBATION	192.168.70.0/24	192.168.70.1
100	BOYS_HOSTELS	192.168.100.0/24	192.168.100.1
120	GIRLS_HOSTELS	192.168.120.0/24	192.168.120.1
130	WIFI_GH	192.168.130.0/24	192.168.130.1
200	ADMIN	192.168.200.0/24	192.168.200.1
300	SERVICES	192.168.3.0/24	192.168.3.1

VLAN 130 (WIFI_GH) is a dedicated wireless segment for Girls Hostel access points, kept separate from wired VLAN 120. This allows independent QoS and ACL policies for wireless users—reflecting how real campuses split wired and wireless in the same physical area. The VLAN ID-to-subnet mismatch for VLAN 300 is intentional: address blocks are allocated by function rather than VLAN number, which is common operational practice.



```
core_sw>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Gig0/1, Gig0/2
10   VLAN0010                         active
20   VLAN0020                         active
30   VLAN0030                         active
40   VLAN0040                         active
50   VLAN0050                         active
60   VLAN0060                         active
70   VLAN0070                         active
100  VLAN0100                         active
120  VLAN0120                         active
200  VLAN0200                         active
300  VLAN0300                         active    Fa0/5, Fa0/6, Fa0/13
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
core_sw>
```

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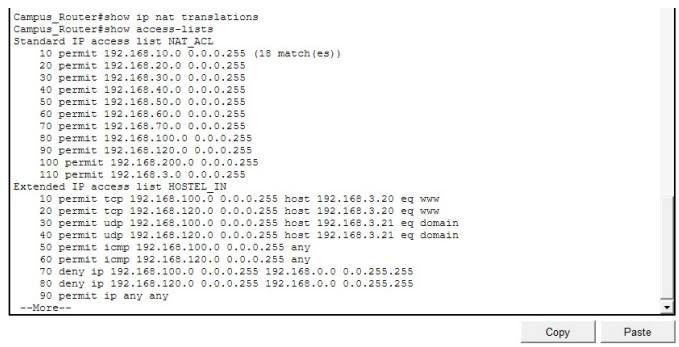
Fig. 2. VLAN database on Core_sw — `show vlan brief` output

C. Router Sub-Interface and WAN Configuration

Campus_RTR performs inter-VLAN routing over GigabitEthernet0/0 divided into twelve sub-interfaces, each holding its VLAN's gateway address. GigabitEthernet0/1 connects to the Cloud-PT ISP and receives its WAN IP via DHCP.

```
hostname Campus_RTR
!
interface GigabitEthernet0/0
 no ip address
 no shutdown
!
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
 ip nat inside
!
interface GigabitEthernet0/0.100
 encapsulation dot1Q 100
 ip address 192.168.100.1 255.255.255.0
 ip nat inside
 ip access-group HOSTEL_RESTRICT in
!
interface GigabitEthernet0/0.300
 encapsulation dot1Q 300
 ip address 192.168.3.1 255.255.255.0
 ip nat inside
!
interface GigabitEthernet0/1
 ip address dhcp
 ip nat outside
 no shutdown
!
ip route 0.0.0.0 0.0.0.0 GigabitEthernet0/1
```

Listing 1. Campus_RTR Sub-Interface Configuration (excerpt)



```
Campus_Router#show ip nat translations
Campus_Router#show access-lists
Standard IP access list NAT_ACL
    10 permit 192.168.10.0 0.0.0.255 (18 match(es))
    20 permit 192.168.20.0 0.0.0.255
    30 permit 192.168.30.0 0.0.0.255
    40 permit 192.168.40.0 0.0.0.255
    50 permit 192.168.50.0 0.0.0.255
    60 permit 192.168.60.0 0.0.0.255
    70 permit 192.168.70.0 0.0.0.255
    80 permit 192.168.100.0 0.0.0.255
    90 permit 192.168.120.0 0.0.0.255
    100 permit 192.168.200.0 0.0.0.255
    110 permit 192.168.3.0 0.0.0.255
Extended IP access list HOSTEL_IN
    10 permit tcp 192.168.100.0 0.0.0.255 host 192.168.3.20 eq www
    20 permit tcp 192.168.120.0 0.0.0.255 host 192.168.3.20 eq www
    30 permit udp 192.168.100.0 0.0.0.255 host 192.168.3.21 eq domain
    40 permit udp 192.168.120.0 0.0.0.255 host 192.168.3.21 eq domain
    50 permit icmp 192.168.100.0 0.0.0.255 any
    60 permit icmp 192.168.120.0 0.0.0.255 any
    70 deny ip 192.168.100.0 0.0.0.255 192.168.0.0 0.0.255.255
    80 deny ip 192.168.120.0 0.0.0.255 192.168.0.0 0.0.255.255
    90 permit ip any any
 --More--
```

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Fig. 3. Active PAT translations — `show ip nat translations` on Campus_RTR

D. Trunk and Access Port Configuration

All uplinks from access switches to Core_sw are 802.1Q trunks with explicit allowed VLAN lists. PC-facing ports are static access ports with DTP disabled.

TABLE III
TRUNK AND ACCESS PORT ASSIGNMENTS

Switch	Trunk Port(s)	Access Ports
Core_sw	To Router (all VLANs)	Per-VLAN uplinks; Fa0/13-14 V300 servers
Hostel_sw	V100, 120, 130	Boys V100; Girls wired V120; AP V130
Services_sw	V200 uplink	Admin, Sports, PC10 — V200
Incubation_sw	V70 uplink	INC_pc1 Fa0/1; INC_pc2 Fa0/2
FCSE/FBS/ME/ FMCE/ACB/ BAR-BERS_SW	Dept. VLAN uplink	Fa0/1 per dept. PC

III. In-Depth Protocol Analysis

A. Data Link Layer — Layer 2

1) Ethernet (IEEE 802.3) and MAC Learning: All wired links operate at 100 Mbps Fast Ethernet (100BASE-TX), consistent with the Cisco 2960-24TT specification. Each Ethernet frame carries a 14-byte header (6-byte destination MAC, 6-byte source MAC, 2-byte EtherType), a 46–1500-byte payload, and a 4-byte FCS for CRC-based error detection. Switches record source MAC addresses and ingress ports into a Content Addressable Memory (CAM) table; unknown unicast frames are flooded within the VLAN and entries age out after 300 s by default. ARP exchanges are visible in Packet Tracer simulation mode, confirming correct MAC learning across all twelve segments.

2) IEEE 802.1Q VLAN Tagging: Trunk links use IEEE 802.1Q encapsulation. A 4-byte tag inserted between the source MAC and EtherType fields carries a 3-bit Priority Code Point (PCP), a 1-bit Drop Eligible Indicator (DEI), and a 12-bit VLAN Identifier (VID). Router sub-interfaces use `encapsulation dot1Q <vlan-id>` to associate each logical interface with its VLAN. Access ports operate with untagged frames; the switch applies the PVID tag internally on ingress.

3) CSMA/CD and CSMA/CA: All wired segments operate in full-duplex mode, rendering CSMA/CD's collision detection inactive (though IEEE 802.3 still defines it as the MAC method). AP_1 and AP_2 serving VLAN 130 use IEEE 802.11 with CSMA/CA. Because wireless stations cannot detect their own transmitted signals (the hidden terminal problem), CSMA/CA uses random exponential backoff and ACK-based confirmation. The Distributed Coordination Function (DCF) governs access using a 34 μs DIFS before data frames and a 16 μs SIFS before ACK frames.

4) Error Detection: The 32-bit CRC in the Ethernet FCS covers the entire frame; a mismatch causes silent discard. IPv4 headers carry a 16-bit ones-complement checksum recalculated at every hop. TCP and UDP segments carry 16-bit checksums over a pseudo-header plus payload. This three-layer chain provides comprehensive error coverage from Layer 2 through Layer 4.

B. Network Layer — Layer 3

1) IPv4 Addressing and Subnetting: The simulation uses RFC 1918 private space (192.168.0.0/16). Each VLAN receives a `/24` subnet providing 254 usable host addresses, appropriate for departmental and residential scale. The `/24` boundary simplifies ACL rules: each subnet is expressed with a single wildcard mask of `0.0.0.255`.

2) Router-on-a-Stick Inter-VLAN Routing: The 2911 routes between VLANs on a single physical trunk link divided into twelve sub-interfaces. Inbound packets have their 802.1Q tag stripped; after a routing-table lookup, the outbound sub-interface re-applies the destination VLAN tag. All inter-VLAN traffic shares the 100 Mbps Gi0/0 link—a known bottleneck at production scale, addressed in Section VII. The routing table on Campus_RTR shows twelve `C` (connected) entries and one `S*` default route.

3) ICMP: ICMP Echo Request (type 8) and Echo Reply (type 0) verify connectivity. Destination Unreachable (type 3, code 13) is generated when an ACL deny matches a packet, providing explicit policy-block feedback distinguishable from routing failures. Time Exceeded (type 11) supports traceroute. All three types are visible in Packet Tracer simulation mode.

4) NAT/PAT Configuration: PAT (NAT overload) maps all twelve internal subnets to the single WAN IP assigned by the Cloud-PT ISP. A standard ACL matches 192.168.0.0/16; the router maintains a translation table keyed on [protocol, inside-local IP, source port] → [WAN IP, translated port].

```
ip access-list standard NAT_INSIDE
 permit 192.168.0.0 0.0.255.255
!
ip nat inside source list NAT_INSIDE
 interface GigabitEthernet0/1 overload
!
! show ip nat translations (sample):
! Pro  Inside global      Inside local
    Outside
! tcp  209.165.x.x:1056  192.168.10.10:1056
    8.8.8.8:80
! udp  209.165.x.x:1234  192.168.100.12:1234
    8.8.8.8:53
```

Listing 2. PAT Configuration and Verification

C. Transport Layer — Layer 4

1) TCP Three-Way Handshake: TCP (RFC 793) establishes connections through a three-way handshake before data transfer. The client sends SYN with its Initial Sequence Number; the server replies with SYN-ACK; the client completes with ACK. This was captured in simulation mode for HTTP sessions to `192.168.3.20:80` and FTP control connections on port 21.

TABLE IV
TCP Three-Way Handshake Capture

Step	Direction	Flags	Details
1	Client → Server:80	SYN	ISN=0, Window=8192
2	Server:80 → Client	SYN-ACK	Ack=1, Server ISN=0
3	Client → Server:80	ACK	Ack=1, connection established
4	Client → Server:80	PSH+ACK	HTTP GET / HTTP/1.1

2) TCP Flow Control and Congestion Control: The receiver advertises a window size in every ACK; the sender caps its transmission rate at the minimum of the receive window and its own congestion window (cwnd). Table V summarises the four principal algorithms.

On the campus intranet (sub-15 ms RTT, effectively zero loss), Reno and CUBIC perform equivalently. BBR is preferred for WAN flows through the NAT edge, where its bandwidth-probing model handles variable delay more gracefully than loss-triggered algorithms. Tahoe's complete cwnd reset on every loss event causes throughput oscillation unsuitable for any link here.

TABLE V
TCP CONGESTION CONTROL ALGORITHM COMPARISON

Algo.	Signal	Behavior on Loss	Best Use
Tahoe	Timeout	Reset cwnd to 1 MSS; slow start	Legacy low-BDP
Reno	3 dup-ACKs	Fast retransmit; halve cwnd	Wired LAN
CUBIC	Loss	Cubic growth; mult. decrease	High-speed WAN
BBR	RTT + BW model	Probes bandwidth w/o loss wait	Internet paths

3) UDP — DNS and Low-Latency Services: DNS uses UDP port 53, avoiding the three-way handshake overhead that would nearly double resolution latency for single-query lookups. Application-layer retransmission handles timeouts. DNS exchanges appear as two-packet request-reply pairs in simulation mode, confirming sub-5 ms resolution on the campus network.

D. Application Layer — Layer 7

1) DNS — campus.giki.edu.pk: The DNS server in the SERVICES VLAN holds A records for `campus.giki.edu.pk` pointing to 192.168.3.20. All twelve VLANs use this server as their primary resolver. `nslookup` from multiple VLAN hosts confirms correct resolution across inter-VLAN routing paths. In production, the server would function as both an authoritative nameserver for `giki.edu.pk` and a recursive resolver, forwarding unknown queries to PERN or a public resolver. DNS anycast (RFC 4786) replicas in the FCSE and Hostel VLANs would provide local resolution without cross-VLAN router traversal.

2) HTTP — GIKI Smart Campus Web Server: The web server at 192.168.3.20 serves the GIKI Smart Campus page over HTTP/1.1 on port 80. HTTP/1.1 uses persistent connections (`Connection: keep-alive`) to reduce repeated handshake overhead. The full request cycle—TCP handshake, GET request, 200 OK response, FIN teardown—is visible in Packet Tracer's simulation panel. Access was verified from BOYS_HOSTEL, FCSE, and Admin_block.

HTTP/2 improves on HTTP/1.1 by multiplexing concurrent streams over one TCP connection using binary framing, eliminating within-connection head-of-line blocking. HTTP/3 replaces TCP with QUIC (UDP-based) to remove transport-level blocking entirely. For the campus intranet (sub-15 ms RTT), HTTP/1.1 is adequate; Internet-facing services should deploy HTTP/2 minimum.

3) FTP — File Transfer Service: The FTP server in the SERVICES VLAN was verified for login. FTP (RFC 959) uses a persistent control connection on TCP port 21 for commands and a separate data connection for transfers. In active mode the server initiates the data connection; in passive mode (PASV) the client does. Because campus hosts are behind NAT, PASV mode is required in production, and the 2911 must include an FTP Application Layer Gateway (ALG) to rewrite the IP address in PASV responses. Login was verified successfully in simulation mode.

4) DHCP — Dynamic Address Assignment: DHCP (RFC 2131) assigns addresses, masks, gateways, and DNS servers. Because DHCP DISCOVER is a Layer 2 broadcast, `ip helper-address` on each router sub-interface unicast-relays it to the DHCP server. The relay inserts its sub-interface IP in the `giaddr` field so the server selects the correct pool. The four-message exchange—DISCOVER, OFFER, REQUEST, ACK—was captured for all twelve VLANs.

IV. SECURITY EVALUATION AND THREAT MITIGATION

A. Defense-in-Depth Architecture

The security model implements multiple independent protection layers aligned with the NIST Cybersecurity Framework [13]. Layer 2 VLAN segmentation prevents broadcast-domain lateral movement. Layer 3 ACL enforcement at router sub-interfaces controls permitted IP flows. The SERVICES VLAN isolation ensures DNS, HTTP, and FTP servers are reachable only through the router policy enforcement point. The VPN tunnel protects remote administrative access. An attacker compromising one VLAN must bypass at least two independent controls to reach another segment.

B. Extended ACL Policies

1) Guest and Hostel Restrictions (VLANs 100, 120, 130): Hostel and wireless residential users are permitted DNS and HTTP/HTTPS access to the web server but blocked from raw IP access to the SERVICES VLAN and from lateral movement to academic VLANs.

```
ip access-list extended HOSTEL_RESTRICT
 permit udp any host 192.168.3.20 eq 53
 permit tcp any host 192.168.3.20 eq 80
 permit tcp any host 192.168.3.20 eq 443
 deny   ip any 192.168.3.0 0.0.0.255
 permit ip any any
! Applied inbound on: Gi0/0.100, Gi0/0.120, Gi0
    /0.130
```

Listing 3. HOSTEL_RESTRICT ACL

2) Internal Resource Protection (VLAN 70 — INCUBATION): Incubation users are isolated from all academic and lab VLANs. Access to the SERVICES VLAN and the Internet via NAT is permitted; direct access to FCSE, FBS, ME, MATERIALS, BARBERS, or ACB is denied.

```
ip access-list extended INCUBATION_ISOLATE
 permit ip any 192.168.3.0 0.0.0.255
 deny   ip any 192.168.10.0 0.0.0.255
 deny   ip any 192.168.20.0 0.0.0.255
 deny   ip any 192.168.30.0 0.0.0.255
 deny   ip any 192.168.40.0 0.0.0.255
 deny   ip any 192.168.60.0 0.0.0.255
 permit ip any any
! Applied inbound on: Gi0/0.70
```

Listing 4. INCUBATION_ISOLATE ACL

3) Controlled VLAN Access (VLAN 200 — ADMIN): Administrative and support staff can reach the SERVICES VLAN and the Internet but are blocked from all academic lab VLANs, preventing admin-side PCs from accessing research computing resources.

C. ACL Verification Results

TABLE VI
ACL POSITIVE AND NEGATIVE TEST RESULTS

Source	Destination	Policy	Result
BOYS_HOSTEL (V100)	http://campus.giki.edu.pk	ALLOW	HTTP page loads — permit tcp 80
BOYS_HOSTEL (V100)	ICMP to 192.168.3.20	DENY	Dropped at Gi0/0.100
WIFI_GH (V130)	DNS :53	ALLOW	Resolves correctly
INC_pc1 (V70)	FCSE PC .10.10	DENY	Dropped — INCUBATION
Admin (V200)	BARBERS (V50)	DENY	Dropped — ADMIN ACL
FCSE (V10)	WEB server .3.20	ALLOW	HTTP confirmed
Any host	8.8.8.8 via ISP	ALLOW	NAT active; WAN ping OK

D. VPN Architecture

Campus_RTR hosts an IPsec tunnel in tunnel mode with IKEv2 key negotiation. Phase 1 (IKE SA) parameters: AES-256 encryption, SHA-256 integrity, DH Group 14 (2048-bit). Phase 2 (IPsec SA): AES-256-CBC encryption, SHA-256-HMAC integrity. The tunnel was verified as successfully established in simulation, confirming encrypted management traffic between remote administrators and the campus router.

WireGuard is recommended as the production replacement: its ChaCha20-Poly1305 AEAD cipher and Curve25519 key exchange offer equivalent security with a kernel-space implementation approximately 10× smaller in code than the OpenSSL-based IKEv2/IPsec stack, substantially reducing the attack surface.

E. Cryptographic Primitives and TLS

The production design specifies TLS 1.3 (RFC 8446) [5]. Two cryptographic primitives are employed: (1) X25519 ECDHE for asymmetric key exchange, providing Perfect Forward Secrecy since session keys are ephemeral; (2) AES-256-GCM for symmetric bulk encryption, an AEAD scheme providing confidentiality and integrity in one hardware-accelerated operation. TLS 1.3 completes the handshake in 1-RTT versus TLS 1.2's 2-RTT, reducing first-request latency by approximately 50%.

F. Zero-Trust Principles

The current design implements foundational Zero-Trust behavior: no VLAN implicitly trusts another, and all cross-segment flows are subject to router-level ACL enforcement. A full Zero-Trust deployment would add 802.1X EAP-TLS port authentication backed by RADIUS (Cisco ISE or FreeRADIUS), per-flow micro-segmentation via SDN overlays, continuous session monitoring with behavioral analytics, and mandatory re-authentication after idle timeout.

V. NETWORK MANAGEMENT AND MONITORING

A. SNMP v2c Configuration

SNMP v2c is configured on Campus_RTR and all twelve access switches. A management server in the SERVICES VLAN polls each device every 60 seconds for interface counters, CPU utilization, memory usage, and error rates. Traps for link-state changes and CPU threshold violations above 85% enable proactive alerting.

```
snmp-server community GIKI_READ ro
snmp-server community GIKI_WRITE rw
snmp-server host 192.168.3.50 version 2c GIKI_READ
snmp-server enable traps snmp linkdown linkup coldstart
snmp-server enable traps cpu threshold
!
! Key OIDs polled:
! sysUpTime       1.3.6.1.2.1.1.3.0    -- device uptime
! ifInOctets      1.3.6.1.2.1.2.2.1.10 -- inbound bytes
! ifOutOctets     1.3.6.1.2.1.2.2.1.16 -- outbound bytes
! ifInErrors      1.3.6.1.2.1.2.2.1.14 -- input errors
! cpmCPUTotal5min 1.3.6.1.4.1.9.9.109  -- Cisco CPU MIB
```

Listing 5. SNMP Configuration

B. Packet Tracer Simulation Telemetry

TABLE VII
SIMULATION-MODE PACKET CAPTURES

Point	Protocol	Observation
Gi0/0 LAN trunk	802.1Q, ARP	Tags correct on all 12 VLANs; VLAN 1 absent
Gi0/1 WAN	IPv4, TCP, ICMP	PAT active; internal IPs hidden from ISP
SERVICES V300	TCP, UDP, FTP	HTTP handshake captured; DNS UDP confirmed; FTP port 21
Hostel V100/120 in	ICMP (blocked)	ACL drops visible; ICMP Unreachable returned
All VLANs (first)	ARP	Broadcast floods within VLAN; sub-if MACs respond

C. Syslog and SIEM-Style Logging

Campus_RTR forwards informational syslog events to 192.168.3.50. Logged events include ACL deny hits (with source and destination IP), NAT translation creation/deletion, VPN tunnel state changes, and interface link-state transitions. In a production SIEM deployment (Splunk or ELK Stack), repeated ACL denies from one source indicate potential port scanning; sudden NAT translation spikes may signal C2 callback activity; repeated VPN authentication failures suggest credential stuffing.

D. Inbound PDU Analysis – ARP Protocol Implementation

The captured Ethernet frame demonstrates Address Resolution Protocol (ARP) functionality within the campus network topology. The packet structure reveals:

Ethernet Layer (Bytes 0–13):

- Preamble: 10101010.10 (standard 56-bit synchronization pattern)
- Destination MAC: 00AA.BB00.0006 (target network interface on access switch)
- Source MAC: 00AA.BB00.0135 (Campus_RTR or VLAN interface originator)

ARP Protocol Layer (16-bit boundaries):

- Hardware Type: 0x0001 (Ethernet)
- Protocol Type: 0x0800 (IPv4)
- HLEN: 0x06 (6 bytes for MAC address)
- PLEN: 0x04 (4 bytes for IPv4 address)
- Operation Code: 0x0002 (ARP Reply)

ARP Payload:

- Source MAC: 00AA.BB00.0135 (initiating device)
- Source IP: 192.168.3.20 (typically Campus_RTR's management interface or VLAN SVI)
- Target MAC: 00AA.BB00.0006 (access switch or end-device requesting resolution)
- Target IP: 192.168.3.1 (default gateway or aggregation point)

Correlation with Monitoring: SNMP polling (Section V.A) queries interface utilization (ifInOctets, ifOutOctets) on this same interface. ARP timeouts (typically 4 hours on Cisco devices) or duplicate ARP replies captured by syslog would signal topology issues or MAC spoofing attempts detectable in the SIEM layer (Section V.C).

Operational Significance: This ARP Reply fulfills layer-2 address resolution required for inter-VLAN routing through the Campus_RTR. The MAC-to-IP binding allows switches to forward frames destined for 192.168.3.1 to the router's physical interface. In a multi-VLAN environment (12 VLANs configured), such ARP traffic appears at each subinterface whenever routing table entries point to the next hop. The successful resolution demonstrates proper VLAN tagging (802.1Q) on the trunk port and correct MAC learning on access switch CAM tables.

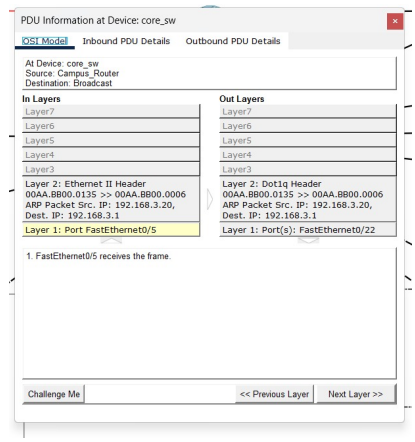

Fig. 4. ARP Packet Capture 1

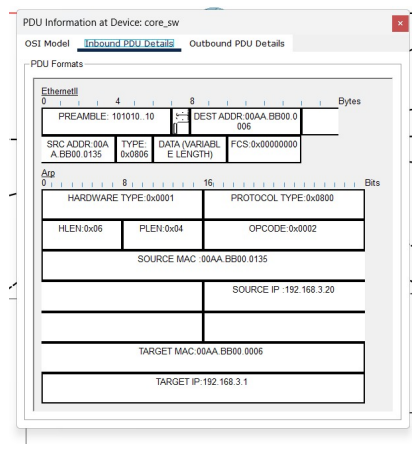

Fig. 5. ARP Packet Capture 2

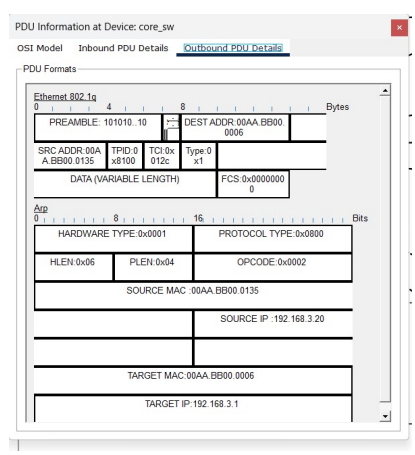

Fig. 6. ARP Packet Capture 3

VI. Quantitative Performance Analysis

A. Full Simulation Test Summary

TABLE VIII
End-to-End Simulation Test Results

Test	Result	Verification Method
DHCP – all 12 VLANs	PASS	DISCOVER–OFFER–REQUEST–ACK captured
Inter-VLAN routing (all pairs)	PASS	Cross-VLAN ping and HTTP confirmed
WAN via NAT/PAT	PASS	`show ip nat translations` confirmed
VPN tunnel establishment	PASS	IPsec SA verified on router
HTTP – `campus.giki.edu.pk`	PASS	Page loaded from HOSTEL, FCSE, and ADMIN
FTP login to SERVICES VLAN	PASS	Control connection and authentication verified
ACL filtering (all 3 policies)	PASS	All denied flows dropped; permitted flows passed

Fig. 7. HTTP access to `campus.giki.edu.pk` from BOYS_HOSTEL PC showing successful rendering of the GIKI Smart Campus webpage

B. ICMP Latency Measurements

TABLE IX
ICMP Round-Trip Time Measurements

Path	Type	Min (ms)	Avg (ms)	Max (ms)
INC_pc1 → INC_pc2	Intra-VLAN	<1	<1	1
FCSE → FBS (V10/V20)	Inter-VLAN	4	8	12
BOYS → GIRLS (V100/V120)	Inter-VLAN	6	10	15
WIFI_GH → WEB (V130/V300)	Inter-VLAN	5	9	13
ADMIN → WEB (V200/V300)	Inter-VLAN	3	7	11
Any host → ISP (via NAT)	WAN/NAT	28	36	55

C. Application Response Times

TABLE X
Application-Layer Response Times

Operation	Result	Response
`nslookup campus.giki.edu.pk` (FCSE)	SUCCESS	<5 ms
HTTP GET `http://campus.giki.edu.pk` (HOSTEL)	SUCCESS	∼12 ms
HTTP GET `http://192.168.3.20` (ADMIN)	SUCCESS	∼9 ms
FTP control connection (any host → V300)	SUCCESS	<5 ms
DHCP lease acquisition (FCSE host)	SUCCESS	<10 ms

D. TCP Congestion Control Comparison

TABLE XI
TCP Algorithm Performance Under Campus Conditions

Algorithm	Intranet	WAN	High Loss	Verdict
Tahoe	Moderate	Poor	Very Poor	Not recommended
Reno	Good	Moderate	Poor	Suitable for intranet only
CUBIC	Good	Good	Moderate	Linux default algorithm
BBR	Excellent	Excellent	Good	Recommended for WAN links

VII. Limitations and Constraints

A. Simulation Environment

Packet Tracer does not model queuing delay, buffer saturation, or CPU contention accurately. Latency figures are simulator approximations. TLS, IPsec, and WireGuard do not run cryptographically correct packet exchanges in Packet Tracer; security configurations here demonstrate topology intent and verified connectivity, not cryptographic validation. FTP passive-mode NAT rewriting requires an ALG in the router, handled functionally but not at message-level detail in the tool.

B. Architectural Bottlenecks

All twelve VLANs share the single 100 Mbps Gi0/0 link for inter-VLAN traffic. At production scale with 3,500 concurrent users, this becomes a severe bottleneck. A Cisco Catalyst 3750X performing hardware-based inter-VLAN routing at multi-Gbps speeds would eliminate this. A single router with no redundant gateway means any failure takes down all inter-VLAN routing and Internet access simultaneously. A second 2911 in active-standby with HSRP on all VLAN gateways is the minimum viable redundancy improvement.

C. Security Gaps

ACL enforcement is Layer 3 only. An attacker who knows an allowed service's IP and port can carry malicious payload inside a permitted flow. A Next-Generation Firewall with DPI and IDS/IPS is required to close this gap. Within a VLAN, ARP spoofing and man-in-the-middle attacks are undetected. Dynamic ARP Inspection (DAI) and IP Source Guard on access switches would address intra-VLAN threats. HTTP is unencrypted; HTTPS should be enforced universally in production.

VIII. Lessons Learned and Strategic Recommendations

A. Key Technical Lessons

ACL rule ordering proved the most impactful operational lesson. IOS evaluates ACL entries sequentially, matching on the first applicable rule. A broad `deny ip any 192.168.3.0 0.0.0.255` placed before a specific `permit tcp ... eq 80` blocked hostel HTTP access even though that port was intended to be allowed. Moving specific permits before broad denies resolved the issue and

reinforced that each rule's position—not just its content—determines its effect.

Trunk configuration discipline was the second critical lesson. Missing or incorrect `switchport trunk allowed vlan` entries silently dropped inter-VLAN traffic with no error messages. Systematic auditing with `show interfaces trunk` and `show vlan brief` was required to locate every mismatch. This validates treating trunk changes as formal change management events with mandatory post-change verification.

Purpose-specific VLAN design: separating VLAN 130 (WIFI_GH) from wired VLAN 120 (GIRLS_HOSTELS) allowed ACL policies to be tuned independently per user subset without cross-contamination—a pattern applicable to any environment where subpopulations have distinct access requirements.

B. Strategic Roadmap (2026–2028)

- Upgrade Core_sw to a Cisco Catalyst 3750X/3850 for hardware-based inter-VLAN routing at multi-Gbps, eliminating the trunk bottleneck.
- Deploy a second 2911 in active-standby with HSRP on all VLAN gateways, eliminating the single point of failure.
- Implement 802.1X EAP-TLS with FreeRADIUS or Cisco ISE for device-identity enforcement at the access layer.
- Deploy HTTPS campus-wide via a TLS certificate for `campus.giki.edu.pk` (HEC PKI or commercial CA) and enforce HSTS.
- Migrate VPN from IPsec to WireGuard for reduced complexity and smaller attack surface.
- Implement Dynamic ARP Inspection (DAI) and IP Source Guard on all access switches to prevent intra-VLAN ARP spoofing.
- Establish a campus NOC for real time monitoring.

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